

Supplementary Material: Proof of Proposition 1

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Proposition 1. *The frequency nadir feasible region is convex with respect to the inertia H and reserve $R = \{R^x\}_{x \in \mathcal{X}}$.*

Proof. Under the setting $D = 0$, the swing equation can be written as

$$\frac{2H}{f_0} \frac{d\Delta f(\tau)}{d\tau} = \Delta P(\tau) - P^{imb}. \quad (1)$$

With the initial condition $\Delta f(0) = 0$, integrating (1) from 0 to τ gives

$$\Delta f(\tau) = \frac{f_0}{2H} \left[\int_0^\tau \Delta P(s) ds - P^{imb} \tau \right]. \quad (2)$$

The total PFR response is modeled as the sum of heterogeneous ramp responses, i.e.,

$$\Delta P(\tau) = \sum_{x \in \mathcal{X}} \min \left\{ \max \left[0, \frac{R^x}{T^x} (\tau - t^x) \right], R^x \right\}, \quad (3)$$

For each resource type x , define

$$a_x(\tau) = \frac{(\tau - t^x)_+^2 - (\tau - t^x - T^x)_+^2}{2T^x}, \quad \forall x \in \mathcal{X}, \quad (4)$$

where $(\cdot)_+ = \max\{\cdot, 0\}$. Then, the integral of the PFR response can be expressed as

$$\int_0^\tau \Delta P(s) ds = \sum_{x \in \mathcal{X}} R^x a_x(\tau). \quad (5)$$

Substituting (5) into (2), the frequency deviation is obtained as

$$\Delta f(\tau) = \frac{f_0}{2H} \left[\sum_{x \in \mathcal{X}} R^x a_x(\tau) - P^{imb} \tau \right]. \quad (6)$$

The FN constraint requires that the frequency deviation should not be lower than the allowable limit, namely,

$$\Delta f(\tau) \geq -\Delta f^{\max}, \quad \forall \tau \geq 0. \quad (7)$$

Substituting (6) into (7) yields

$$\frac{f_0}{2H} \left[\sum_{x \in \mathcal{X}} R^x a_x(\tau) - P^{imb} \tau \right] \geq -\Delta f^{\max}, \quad \forall \tau \geq 0. \quad (8)$$

Since $H > 0$, (8) is equivalent to

$$P^{imb} \tau \leq \sum_{x \in \mathcal{X}} R^x a_x(\tau) + \frac{2H \Delta f^{\max}}{f_0}, \quad \forall \tau \geq 0. \quad (9)$$

For any given $\tau \geq 0$, $a_x(\tau)$ is a constant determined by the response dead time and reserve delivery time. Hence, (9) is an affine inequality with respect to H and $\mathbf{R} = \{R^x\}_{x \in \mathcal{X}}$. Therefore, for each $\tau \geq 0$, (9) defines a half-space in the $(H, \mathbf{R})$ space.

The FN feasible region is the intersection of these half-spaces over all $\tau \geq 0$. Since the intersection of convex sets is convex, the FN feasible region is convex with respect to H and $\mathbf{R}$. $\square$